

SKILLWEEK9

TASK1: To Implement how to List Active Jobs, Display Status Information, Retrieve Job IDs, Format Output, Update State Information, Test Job Listings.

```
sanjay_dundra@Sanjay:~$ pwd
/home/sanjay_dundra
sanjay_dundra@Sanjay:~$ mkdir -p ~/OS_Lab
cd ~/OS_Lab
sanjay_dundra@Sanjay:~/OS_Lab$ pwd
/home/sanjay_dundra/OS_Lab
sanjay_dundra@Sanjay:~/OS_Lab$ sleep 100 &
[1] 849
sanjay_dundra@Sanjay:~/OS_Lab$ jobs
[1]+  Running                  sleep 100 &
sanjay_dundra@Sanjay:~/OS_Lab$ jobs -l
[1]+  849 Running                  sleep 100 &
sanjay_dundra@Sanjay:~/OS_Lab$ jobs -p
849
sanjay_dundra@Sanjay:~/OS_Lab$ jobs
[1]+  Running                  sleep 100 &
sanjay_dundra@Sanjay:~/OS_Lab$ jobs
[1]+  Done                    sleep 100
```

TASK2: To design how to Identify Target Job, Transfer Terminal Control, Resume Processes, Wait for Completion, Update Job State, Test Foreground Switching.

```
sanjay_dundra@Sanjay:~/OS_Lab$ jobs
sanjay_dundra@Sanjay:~/OS_Lab$ sleep 200
^Z
[1]+  Stopped                  sleep 200
sanjay_dundra@Sanjay:~/OS_Lab$ jobs
[1]+  Stopped                  sleep 200
sanjay_dundra@Sanjay:~/OS_Lab$ bg %1
[1]+  sleep 200 &
sanjay_dundra@Sanjay:~/OS_Lab$ jobs
[1]+  Running                  sleep 200 &
sanjay_dundra@Sanjay:~/OS_Lab$ fg %1
sleep 200
^C
```

TASK3: To Analyze Signal Delivery, Understand Signal Masks, Review Process Groups, Explore Terminal Signals, Test Signal Behavior.


```

Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...

Signal received: 15
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...

Signal received: 15
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Killed
sanjay_dundra@Sanjay: ~/Skill9$

```

```

sanjay_dundra@Sanjay: ~/
sanjay_dundra@Sanjay:~$ cd Skill9
sanjay_dundra@Sanjay:~/Skill9$ ps
  PID TTY          TIME CMD
  738 pts/2    00:00:00 bash
  752 pts/2    00:00:00 ps
sanjay_dundra@Sanjay:~/Skill9$ ps aux | grep signal
root      298  0.0  0.3 112760 25300 ?        Ssl  05:23   0:
00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-up
grade-shutdown --wait-for=signal
sanjay_d+ 727  0.0  0.0   2692  1800 pts/0    S+   05:35   0:
00 ./signal
sanjay_d+ 754  0.0  0.0   4100  2164 pts/2    S+   05:36   0:
00 grep --color=auto signal
sanjay_dundra@Sanjay:~/Skill9$ kill -TERM 727
sanjay_dundra@Sanjay:~/Skill9$ kill -15 727
sanjay_dundra@Sanjay:~/Skill9$ kill -9 727
sanjay_dundra@Sanjay:~/Skill9$ kill -l
 1) SIGHUP       2) SIGINT       3) SIGQUIT      4) SIGILL
 5) SIGTRAP     6) SIGABRT     7) SIGBUS      8) SIGFPE
 9) SIGKILL     10) SIGUSR1    11) SIGSEGV     12) SIGUSR2
13) SIGPIPE     14) SIGALRM    15) SIGTERM     16) SIGSTKFLT
17) SIGCHLD     18) SIGCONT    19) SIGSTOP     20) SIGTSTP
21) SIGTTIN     22) SIGTTOU    23) SIGURG      24) SIGXCPU
25) SIGXFSZ     26) SIGVTALRM 27) SIGPROF     28) SIGWINCH
29) SIGIO       30) SIGPWR     31) SIGSYS      32) SIGRTMIN
33) SIGRTMIN+1  34) SIGRTMIN+2 35) SIGRTMIN+3 36) SIGRTMIN+4
37) SIGRTMIN+5  38) SIGRTMIN+6 39) SIGRTMIN+7 40) SIGRTMIN+8

```

TASK4: To implement how to resume Stopped Jobs, Send Continue Signals, Update Job Status, Maintain Job Table, Verify Execution, Test Background Recovery.

```
sanjay_dundra@Sanjay:~/OS_Lab$ sleep 300 &
[1] 925
sanjay_dundra@Sanjay:~/OS_Lab$ kill -STOP 950
-bash: kill: (950) - No such process
sanjay_dundra@Sanjay:~/OS_Lab$ kill -STOP 925
sanjay_dundra@Sanjay:~/OS_Lab$ jobs
[1]+  Stopped                  sleep 300
sanjay_dundra@Sanjay:~/OS_Lab$ jobs
[1]+  Stopped                  sleep 300
sanjay_dundra@Sanjay:~/OS_Lab$ kill -CONT 925
sanjay_dundra@Sanjay:~/OS_Lab$ jobs
[1]+  Done                    sleep 300
sanjay_dundra@Sanjay:~/OS_Lab$ sleep 300 &
[1] 936
sanjay_dundra@Sanjay:~/OS_Lab$ jobs
[1]+  Running                  sleep 300 &
sanjay_dundra@Sanjay:~/OS_Lab$ jobs -l
[1]+  936 Running              sleep 300 &
sanjay_dundra@Sanjay:~/OS_Lab$ kill -STOP 936
sanjay_dundra@Sanjay:~/OS_Lab$ jobs
[1]+  Stopped                  sleep 300
sanjay_dundra@Sanjay:~/OS_Lab$ kill -CONT 936
sanjay_dundra@Sanjay:~/OS_Lab$ jobs
[1]+  Running                  sleep 300 &
sanjay_dundra@Sanjay:~/OS_Lab$ nano signal.c
[1]+  Done                    sleep 300
```


